

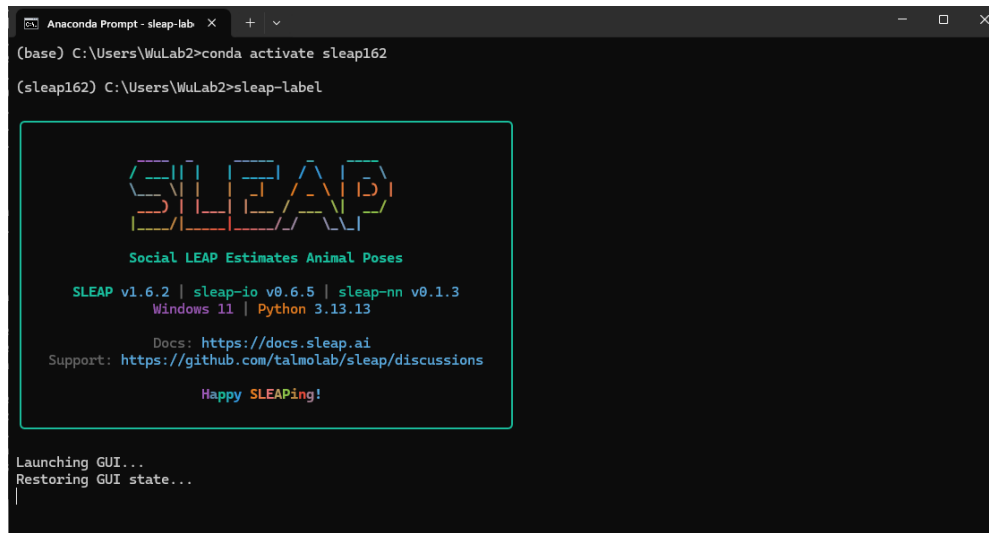
Guide of Sleaf

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1. Open the GUI of Sleap in Anaconda Prompt

```
# bash  
conda activate sleap162  
sleap-label
```



The screenshot shows an Anaconda Prompt window with the title 'Anaconda Prompt - sleap-lab'. The command prompt shows the user activating the 'sleap162' environment and running 'sleap-label'. The output displays the SLEAP logo, version information (SLEAP v1.6.2, sleap-io v0.6.5, sleap-nn v0.1.3), system requirements (Windows 11, Python 3.13.13), documentation and support links, and a 'Happy SLEAPing!' message. At the bottom, it indicates 'Launching GUI...' and 'Restoring GUI state...'.

```
Anaconda Prompt - sleap-lab  
(base) C:\Users\WuLab2>conda activate sleap162  
(sleap162) C:\Users\WuLab2>sleap-label  
  
SLEAP  
Social LEAP Estimates Animal Poses  
  
SLEAP v1.6.2 | sleap-io v0.6.5 | sleap-nn v0.1.3  
Windows 11 | Python 3.13.13  
  
Docs: https://docs.sleap.ai  
Support: https://github.com/talmolab/sleap/discussions  
  
Happy SLEAPing!  
  
Launching GUI...  
Restoring GUI state...
```



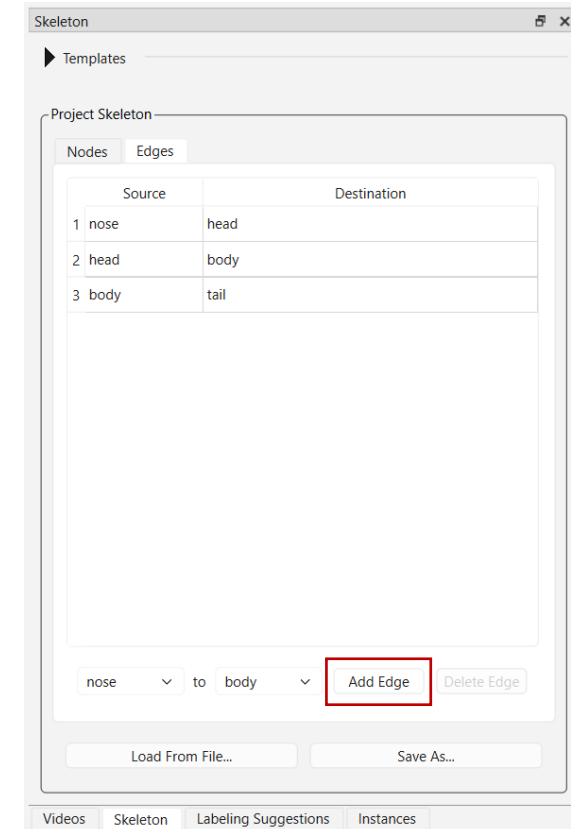
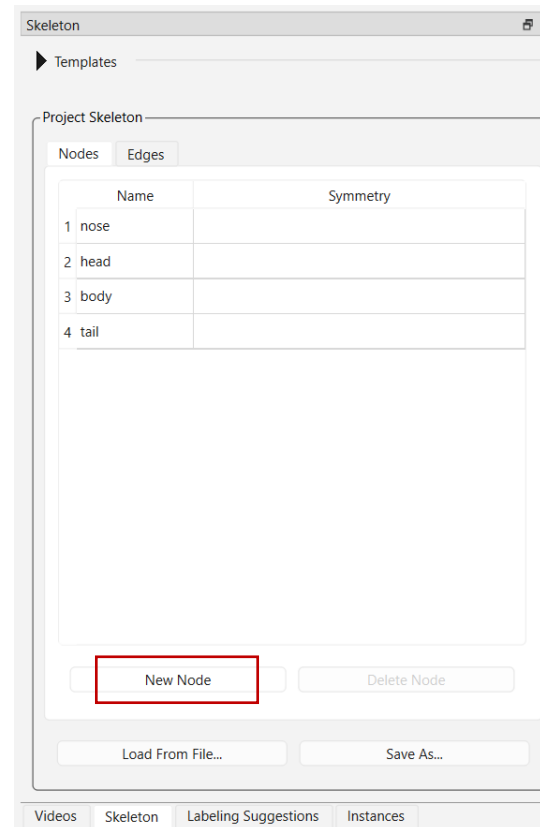
2. Load Videos

Load all videos of a single mouse with “Add Videos” button.



3. Create Skeleton

- Create nodes using “New Node” button.
- Change the name of nodes (for example, nose / head / ...)
- Create edges to connect nodes.



4. Generate Suggested Frames

- Generate 20 suggested frames for each video using “Generate Suggestions” button. (Frame number can be adjusted according to the demand.)

Labeling Suggestions

	Video	Frame	Group	Labeled
1	1: ...	2880	0	1
2	1: ...	5759	0	1
3	1: ...	8638	0	1
4	1: ...	11517	0	1
5	1: ...	14396	0	1
6	1: ...	17275	0	1
7	1: ...	20154	0	1
8	1: ...	23033	0	1
9	1: ...	25912	0	1
10	1: ...	28791	0	1
11	1: ...	31670	0	1

Add current frame

Remove

Clear all

Previous

95/95 labeled (100.0%)

Next

Generate Suggestions

Method: image features

Initial Samples Per Video: 20

Sampling method: stride

Image Scale: 1.00

Compute Features: per video

Image Feature Type: raw images

Brisk Keypoint Threshold: 40

Bag of Features Vocab Size: 20

PCA Components: 5

K-Means Clusters: 5

Samples Per Cluster: 4

Target: all videos

Generate Suggestions

Videos

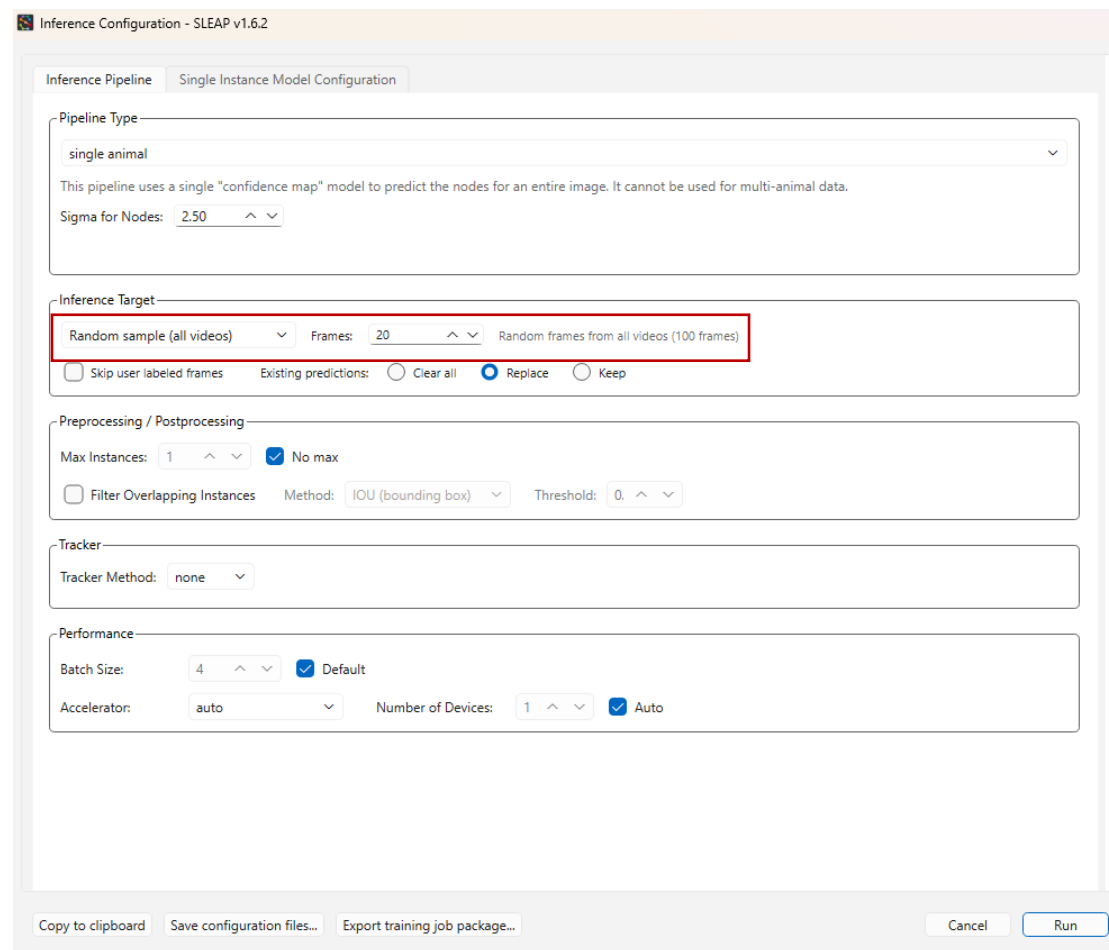
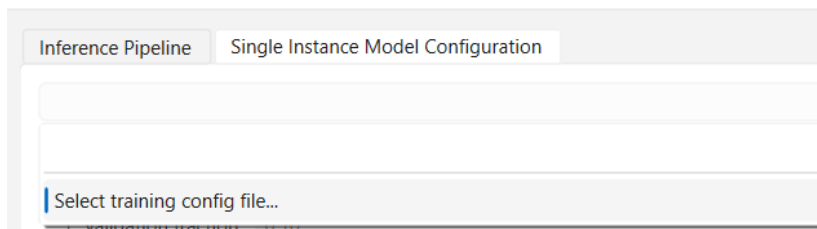
Skeleton

Labeling Suggestions

Instances

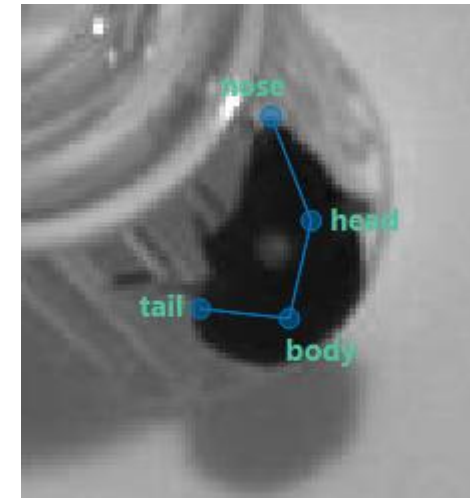
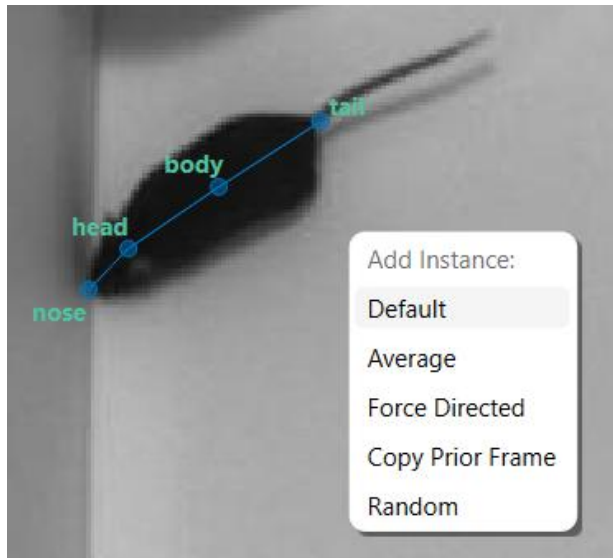
4.2 Generate and Label Frames with a trained model

- If you already had a trained model, you can generate instances with it. (Faster than labeling frames manually.)
- Correct the predicted instances.
- Then go to Step 6 to train the model.



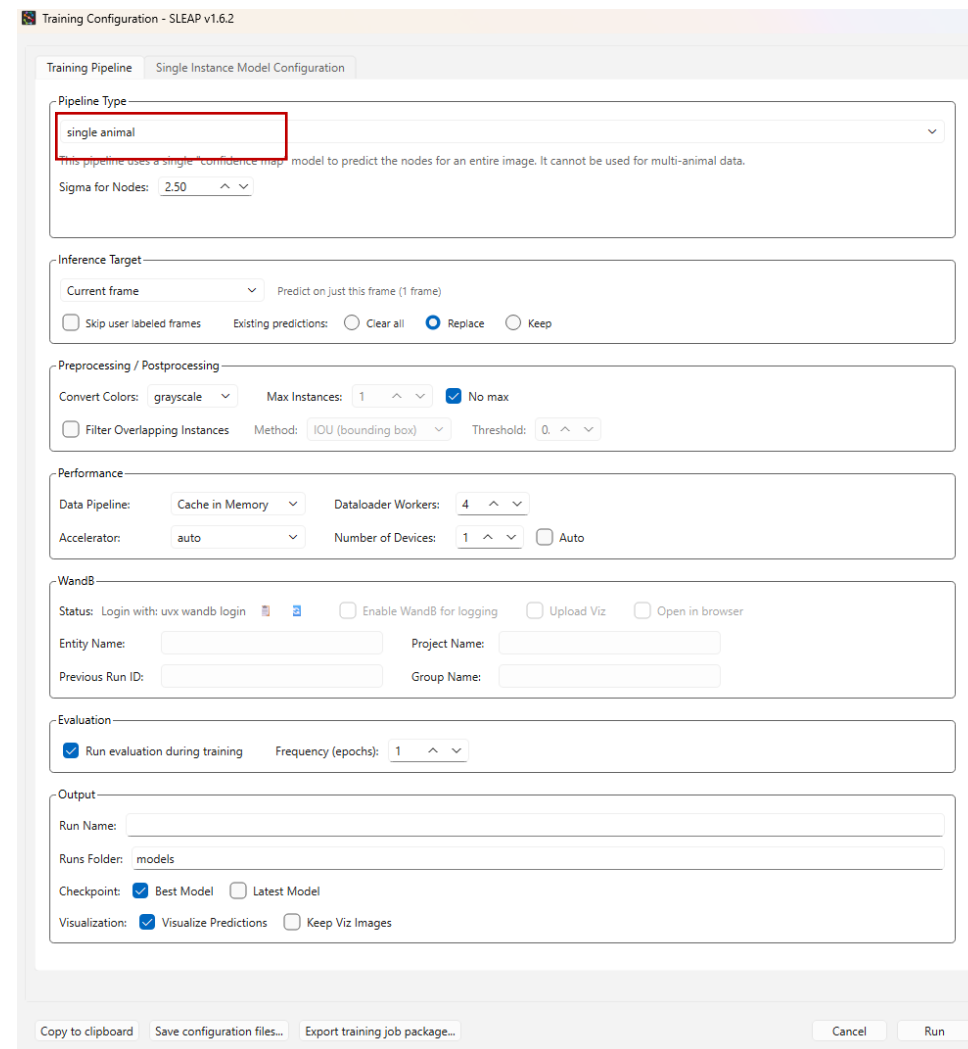
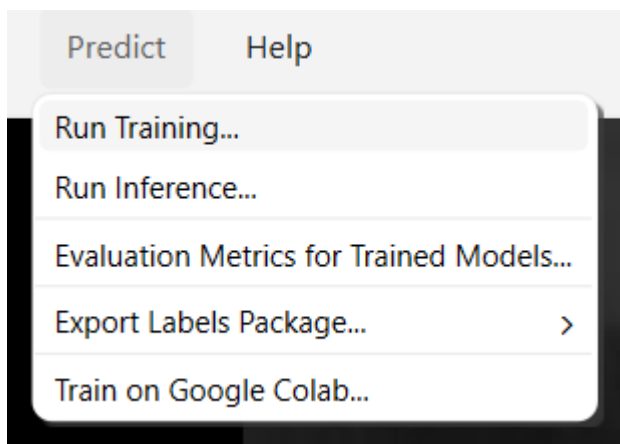
5. Label Frames

- Add instance with right click -> Default, then move each node to an appropriate position.
- Add instances to each individual in the view.
- After labeling all suggested frames, we can check other frames manually and add instance to the frame in which the individual is not easy to label.
- Save the project as “labels.v001.slp”.



6. Run Training

- Choose “single animal”.



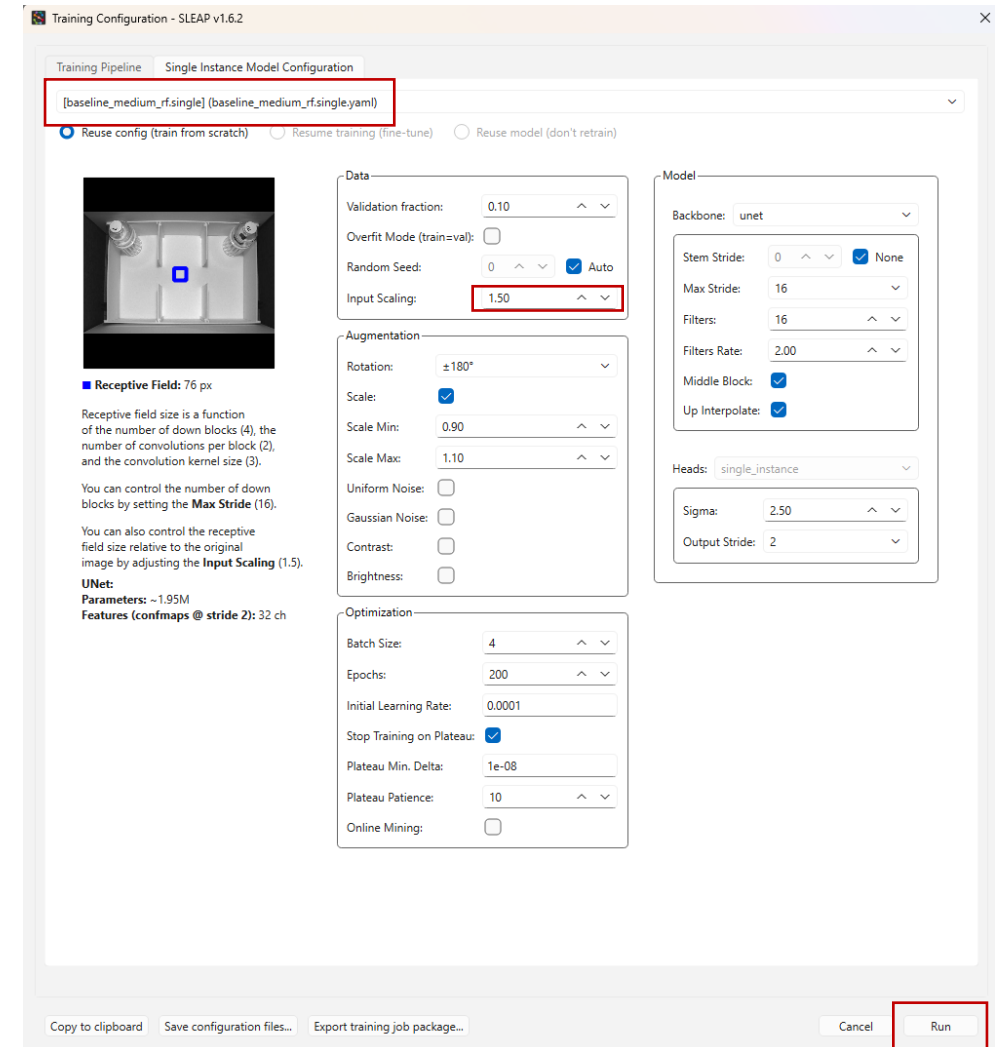
A screenshot of the 'Training Configuration - SLEAP v1.6.2' window. The window is titled 'Training Configuration - SLEAP v1.6.2' and has a close button (X) in the top right corner. It contains several sections for configuring a training pipeline:

- Training Pipeline**: A tab labeled 'Single Instance Model Configuration' is selected. The 'Pipeline Type' dropdown is set to 'single animal', which is highlighted with a red rectangle. Below it, a note states: 'This pipeline uses a single "confidence map" model to predict the nodes for an entire image. It cannot be used for multi-animal data.' The 'Sigma for Nodes' is set to 2.50.
- Inference Target**: The 'Current frame' dropdown is selected. Below it, there are radio buttons for 'Skip user labeled frames', 'Existing predictions: Clear all', 'Replace' (selected), and 'Keep'.
- Preprocessing / Postprocessing**: The 'Convert Colors' dropdown is set to 'grayscale'. 'Max Instances' is set to 1. The 'No max' checkbox is checked. The 'Filter Overlapping Instances' checkbox is unchecked. The 'Method' dropdown is set to 'IOU (bounding box)'. The 'Threshold' is set to 0.
- Performance**: The 'Data Pipeline' dropdown is set to 'Cache in Memory'. 'Dataloader Workers' is set to 4. The 'Accelerator' dropdown is set to 'auto'. 'Number of Devices' is set to 1. The 'Auto' checkbox is unchecked.
- WandB**: The 'Status' is 'Login with: uvx wandb login'. There are checkboxes for 'Enable WandB for logging', 'Upload Viz', and 'Open in browser'. The 'Entity Name', 'Project Name', 'Previous Run ID', and 'Group Name' fields are empty.
- Evaluation**: The 'Run evaluation during training' checkbox is checked. The 'Frequency (epochs)' is set to 1.
- Output**: The 'Run Name' field is empty. The 'Runs Folder' is set to 'models'. The 'Checkpoint' dropdown is set to 'Best Model'. The 'Visualizations' section has checkboxes for 'Visualize Predictions' (checked) and 'Keep Viz Images' (unchecked).

At the bottom of the window, there are buttons for 'Copy to clipboard', 'Save configuration files...', 'Export training job package...', 'Cancel', and 'Run'.

6. Run Training (*continued*)

- Change parameters.
- Click “Run”.



The screenshot shows the 'Training Configuration - SLEAP v1.6.2' window. The 'Single Instance Model Configuration' tab is active. The configuration is based on the '[baseline_medium_rf.single] (baseline_medium_rf.single.yaml)' file. The 'Reuse config (train from scratch)' option is selected. The 'Data' section shows 'Validation fraction: 0.10', 'Overfit Mode (train=val):', 'Random Seed: 0', 'Input Scaling: 1.50', and 'Auto' checked. The 'Augmentation' section shows 'Rotation: ±180°', 'Scale: [checked]', 'Scale Min: 0.90', 'Scale Max: 1.10', 'Uniform Noise: [unchecked]', 'Gaussian Noise: [unchecked]', 'Contrast: [unchecked]', and 'Brightness: [unchecked]'. The 'Optimization' section shows 'Batch Size: 4', 'Epochs: 200', 'Initial Learning Rate: 0.0001', 'Stop Training on Plateau: [checked]', 'Plateau Min. Delta: 1e-08', 'Plateau Patience: 10', and 'Online Mining: [unchecked]'. The 'Model' section shows 'Backbone: unet', 'Stem Stride: 0', 'Max Stride: 16', 'Filters: 16', 'Filters Rate: 2.00', 'Middle Block: [checked]', 'Up Interpolate: [checked]', 'Heads: single_instance', 'Sigma: 2.50', and 'Output Stride: 2'. The 'Receptive Field: 76 px' is displayed. The 'Run' button is highlighted with a red box.

Training Configuration - SLEAP v1.6.2

Training Pipeline Single Instance Model Configuration

[baseline_medium_rf.single] (baseline_medium_rf.single.yaml)

☒ Reuse config (train from scratch) ☐ Resume training (fine-tune) ☐ Reuse model (don't retrain)

Data

Validation fraction: 0.10

Overfit Mode (train=val): ☐

Random Seed: 0 ☒ Auto

Input Scaling: 1.50

Augmentation

Rotation: ±180°

Scale: ☒

Scale Min: 0.90

Scale Max: 1.10

Uniform Noise: ☐

Gaussian Noise: ☐

Contrast: ☐

Brightness: ☐

Optimization

Batch Size: 4

Epochs: 200

Initial Learning Rate: 0.0001

Stop Training on Plateau: ☒

Plateau Min. Delta: 1e-08

Plateau Patience: 10

Online Mining: ☐

Model

Backbone: unet

Stem Stride: 0 ☒ None

Max Stride: 16

Filters: 16

Filters Rate: 2.00

Middle Block: ☒

Up Interpolate: ☒

Heads: single_instance

Sigma: 2.50

Output Stride: 2

Receptive Field: 76 px

Receptive field size is a function of the number of down blocks (4), the number of convolutions per block (2), and the convolution kernel size (3).

You can control the number of down blocks by setting the **Max Stride** (16).

You can also control the receptive field size relative to the original image by adjusting the **Input Scaling** (1.5).

UNet:

Parameters: ~1.95M

Features (confmasks @ stride 2): 32 ch

Copy to clipboard Save configuration files... Export training job package... Cancel Run

6. Run Training (*continued*)

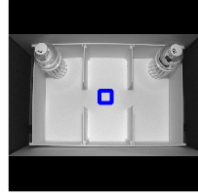
- Or use the trained configuration to train the model.

Training Configuration - SLEAP v1.6.2

Training Pipeline Single Instance Model Configuration

[Trained] 260319_165219_single_instance.n=110(training_config.yaml)

☒ Reuse config (train from scratch) ☐ Resume training (fine-tune) ☐ Reuse model (don't retrain)



■ **Receptive Field:** 76 px

Receptive field size is a function of the number of down blocks (4), the number of convolutions per block (2), and the convolution kernel size (3).

You can control the number of down blocks by setting the **Max Stride** (16).

You can also control the receptive field size relative to the original image by adjusting the **Input Scaling** (1.5).

UNet:
Parameters: ~1.95M
Features (confmasks @ stride 2): 32 ch

Data

Validation fraction: 0.10

Overfit Mode (train=val): ☐

Random Seed: 0 ☒ Auto

Input Scaling: 1.50

Augmentation

Rotation: ±180°

Scale: ☒

Scale Min: 0.90

Scale Max: 1.10

Uniform Noise: ☐

Gaussian Noise: ☐

Contrast: ☐

Brightness: ☐

Optimization

Batch Size: 4

Epochs: 200

Initial Learning Rate: 0.0001

Stop Training on Plateau: ☒

Plateau Min. Delta: 1e-08

Plateau Patience: 10

Online Mining: ☐

Model

Backbone: unet

Stem Stride: 0 ☒ None

Max Stride: 16

Filters: 16

Filters Rate: 2.00

Middle Block: ☒

Up Interpolate: ☒

Heads: single_instance

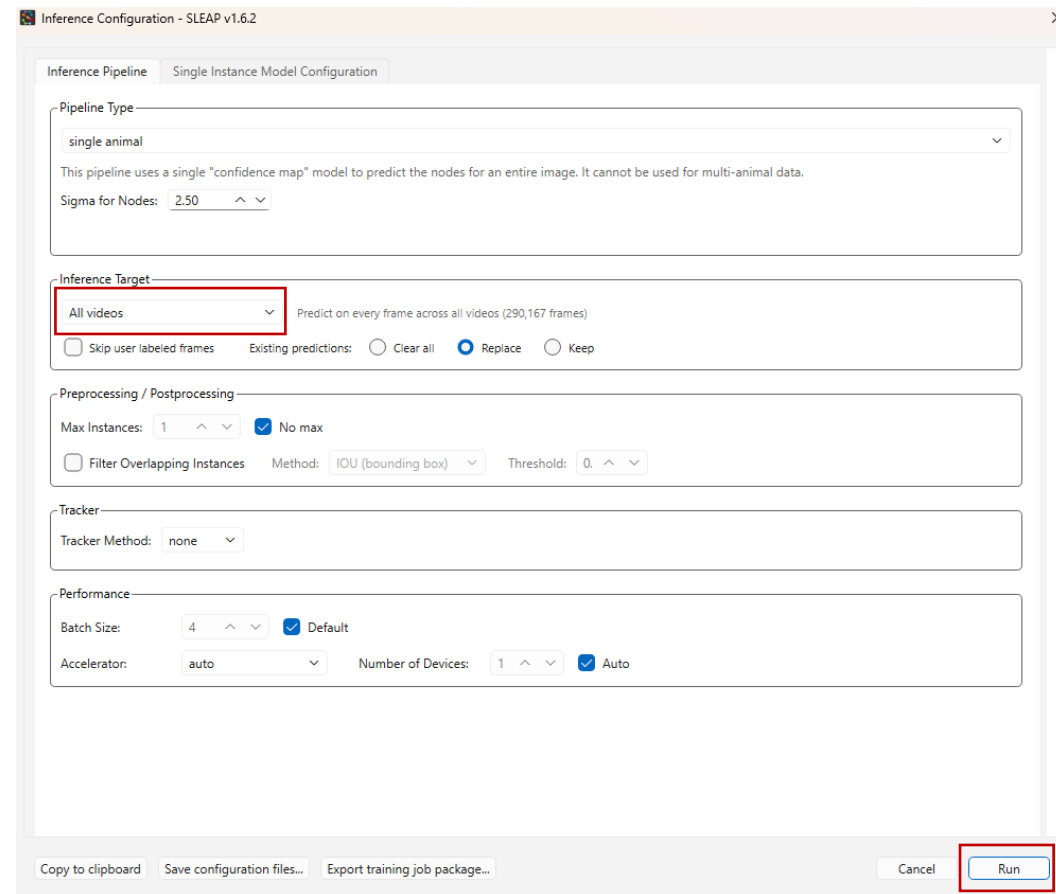
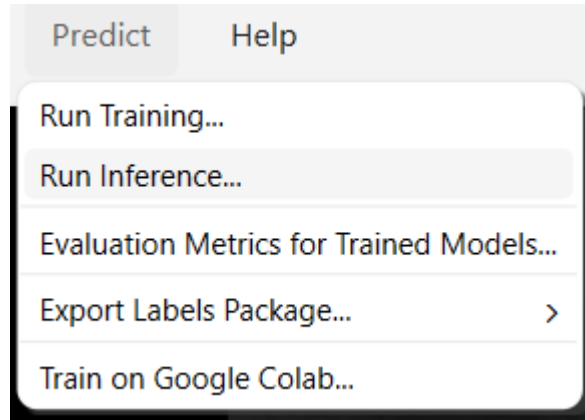
Sigma: 2.50

Output Stride: 2

Copy to clipboard Save configuration files... Export training job package... Cancel **Run**

7. Run Inference

- Use the model trained in 6 to predict on all frames.

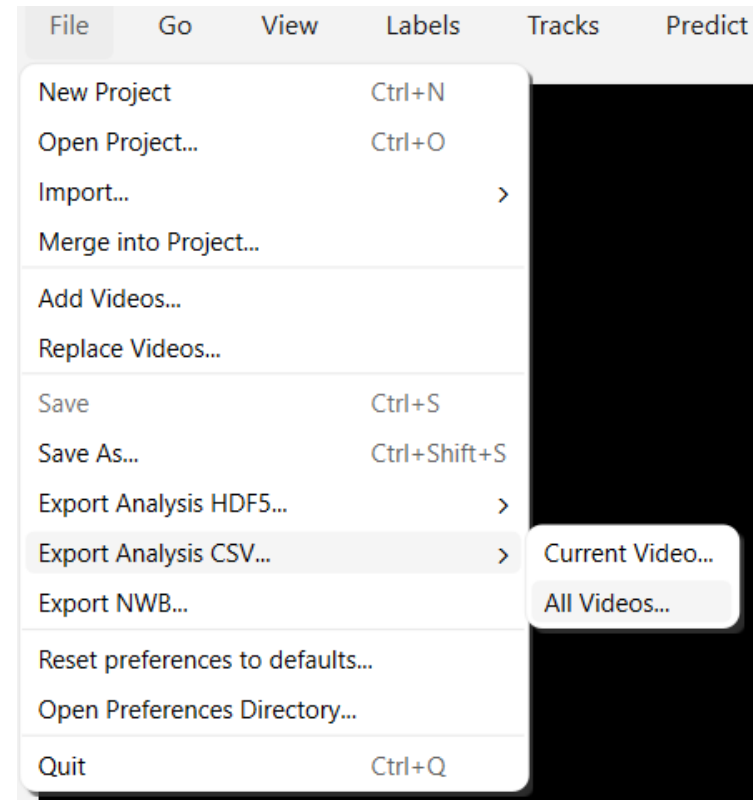


8. Check and Correct Instances

- Check and correct the predicted frames.
- Go to Step 6 to train the model again if the predicted results are not good enough.

9. Export predictions

- Check the predicted frames.
- Export predictions as “csv” files.



Reference

- <https://docs.sleap.ai/dev/>
- T.D. Pereira, N. Tabris, A. Matsliah, D. M. Turner, J. Li, S. Ravindranath, E. S. Papadoyannis, E. Normand, D. S. Deutsch, Z. Y. Wang, G. C. McKenzie-Smith, C. C. Mitelut, M. D. Castro, J. D'Uva, M. Kislin, D. H. Sanes, S. D. Kocher, S. S-H, A. L. Falkner, J. W. Shaevitz, and M. Murthy. SLEAP: A deep learning system for multi-animal pose tracking. Nature Methods, 19(4), 2022.